

ORIGINAL ARTICLE

Evaluation of the effectiveness of dioxidin metaphylaxis of nephrolithiasis in percutaneous nephrolithotripsy

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ABSTRACT

Background and aim. Infectious complications and stone recurrence remain major challenges after percutaneous nephrolithotomy (PCNL) for large kidney stones. This study aimed to evaluate the effectiveness of adjunctive intracavitary 1% dioxidine administration for metaphylaxis of for reducing postoperative infectious complications and stone recurrence following PCNL.

Methods. This prospective, randomized, single-blind controlled trial included 95 patients (aged 18–70 years) with kidney stones >2 cm undergoing PCNL. Patients were randomized 1:1 into control (standard systemic ceftriaxone therapy, n=48) and intervention groups (standard therapy plus intracavitary 10 ml 1% dioxidine intraoperatively and on postoperative days 1–2, n=47). Primary outcomes were bacteriuria and leukocyturia at 7 days and 3 months. Secondary outcomes included Clavien–Dindo complications, stone recurrence at 6 months (low-dose CT), laboratory parameters, and stone composition.

Results. Intracavitary dioxidine significantly reduced bacteriuria at 7 days (10.6% vs 45.8%, $p<0.001$) and 3 months (4.3% vs 52.1%, $p<0.001$). Leukocyturia was markedly lower in the dioxidine group at both time points ($p<0.001$). Clinically significant complications (Clavien–Dindo $\geq$ II) occurred in 4.3% vs 25.0% ($p=0.004$), and stone recurrence at 6 months was 4.3% vs 22.9% ($p=0.008$). Difference-in-Differences analysis confirmed a strong independent effect of dioxidine on urine leukocytes at 1 week (-9.24 ± 2.42 , $p<0.001$) and 3 months (-6.52 ± 2.29 , $p=0.005$). No adverse reactions were observed, and renal function remained stable.



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Conclusions. Adjunctive intracavitary 1% dioxidine during and after PCNL is a safe and effective strategy for reducing postoperative infectious complications and stone recurrence. Larger multicenter trials are warranted. (www.actabiomedica.it)

Key words: percutaneous nephrolithotomy, nephrolithiasis, dioxidine, metaphylaxis, stone recurrence

Introduction

Nephrolithiasis remains one of the most prevalent urological conditions worldwide, with an increasing incidence influenced by dietary, environmental and metabolic risk factors. Current estimates suggest a lifetime prevalence of 10–15% in developed countries, and recurrence rates after surgical treatment can reach up to 50% within 5–10 years if effective meta-phyllaxis strategies are not implemented (1,2). Infectious stones, in particular, poses significant therapeutic challenges, as bacterial colonization of stones contributes not only to the persistence of infection but also to a higher risk of recurrent stone formation (3).

Percutaneous nephrolithotripsy (PCNL) is currently the standard treatment for large and complex renal stones, particularly staghorn calculi and stones >2 cm in diameter. While PCNL offers high stone clearance rates, infectious complications remain a major concern. Postoperative fever occurs in up to 30% of patients, and severe infectious sequelae such as systemic inflammatory response syndrome, septic shock, and even mortality have been reported in 0.3–9.3% of cases (4,5). Established risk factors for sepsis following PCNL include: positive preoperative urine culture, stone composition (struvite, mixed infectious stones), prolonged operative time, high stone burden and multiple access tracts, intraoperative pelvicalyceal pressure elevation, and comorbidities such as diabetes mellitus and chronic pyelonephritis (6).

Given that bacteria within stones and renal tissue may remain undetected despite sterile bladder urine cultures, intraoperative and postoperative antimicrobial strategies are essential for infection control. The limitations of standard systemic antibiotics in preventing bacterial persistence and recurrent stone formation necessitate exploration of adjunctive approaches (1).

Dioxidin (hydroxymethylquinoxalinedioxide) is a broad-spectrum antimicrobial agent with potent bactericidal activity against Gram-negative and some Gram-positive pathogens, including multidrug-resistant strains (7). It has been widely used in Russia and some CIS countries for the treatment of purulent infections, postoperative complications, and urinary tract infections (8). Previous experimental and clinical studies suggest that intracavitary or local administration of dioxidin may provide superior antimicrobial control in surgical fields compared to systemic antibiotics alone (9). However, its role in the meta-phyllaxis of nephrolithiasis following PCNL has not been comprehensively evaluated.

The present study aims to investigate the effectiveness of dioxidin administration in combination with standard antibiotic therapy compared to standard antibiotic therapy alone in patients undergoing PCNL for large renal stones.

Materials and methods

This was a prospective, single-center, randomized, single-blind, controlled clinical trial evaluating the effectiveness of adjunctive intracavitary 1% dioxidine for the metaphylaxis of infectious complications and stone recurrence after percutaneous nephrolithotomy (PCNL). The study was conducted between 2023 and 2025 at the Department of Surgical Diseases No. 2 with Urology, West Kazakhstan Marat Ospanov Medical University, and the Urology Department of Aktobe Medical Center, Aktobe, Kazakhstan. The protocol was approved by the Local Ethics Committee and performed in accordance with the Declaration of Helsinki and CONSORT 2010 guidelines.

Ninety-five adult patients (aged 18–70 years) with kidney stones >2 cm who underwent PCNL were enrolled. Patients were randomized 1:1 into two parallel groups using sequentially numbered opaque sealed envelopes prepared by an independent statistician. Randomization was performed after obtaining written informed consent. The single-blind design applied to patients only.

Inclusion and exclusion criteria Inclusion criteria: informed consent, age 18–70 years, kidney stones >2 cm with density >400 HU, absence of acute genitourinary infection, normal coagulation parameters, and presence of nephrostomy drainage. Exclusion criteria: age <18 or >70 years, pregnancy, coagulopathy, severe musculoskeletal contractures, chronic renal failure, malignancies, urinary tract anomalies, or history of ureteral stent/nephrostomy.

All eligible patients with renal stones >2 cm undergoing PCNL were included irrespective of stone composition. Infection-related characteristics were evaluated using preoperative bladder urine culture, intraoperative renal pelvic urine culture, urinary inflammatory parameters, and postoperative infrared spectroscopy of stone composition. These variables were analyzed to assess their association with postoperative infectious outcomes and stone recurrence.

All patients underwent standard PCNL under general anesthesia with ultrasound-guided access and pneumatic or holmium laser lithotripsy using a 28 Ch nephroscope. In the dioxidine group (n=47), 10 ml of 1% dioxidine solution was administered intracavitarily through the nephrostomy tube intraoperatively and on the 1st and 2nd postoperative days. The control group (n=48) received only standard systemic antibiotic therapy (ceftriaxone) according to the national clinical protocol and antibiogram. Postoperative management was identical in both groups.

Primary outcomes: frequency of bacteriuria and leukocyturia at 7 days and 3 months. **Secondary outcomes:** Clavien–Dindo complications, stone recurrence or residual fragments at 6 months (assessed by low-dose CT), laboratory parameters (hemoglobin, creatinine, blood and urine leukocytes, urine pH), and stone composition (infrared spectroscopy).

Laboratory tests included complete blood count (Sysmex XN-350), urinalysis (DIRUI FUS-2000),

urine culture with EUCAST antibiogram (ADAGIO bio-rad), blood biochemistry (BIOSYSTEM BA-400), and stone chemical composition by infrared spectroscopy (IRSpirit, Shimadzu). Low-dose multi-slice computed tomography (GE Optima 660) was performed before surgery and at 6 months for assessment of residual fragments and recurrence.

Surgical procedure

All patients underwent standard percutaneous nephrolithotomy. The surgical technique was identical in both groups. Operations were performed with pneumatic or holmium laser lithotripter under fluoroscopic control. After lithotripsy and stone extraction, a nephrostomy tube was placed. In the intervention group, 10 ml of 1% dioxidine was injected intracavitarily after completion of lithotripsy.

Statistical analysis

Sample size was calculated using G*Power 3.1.9.7 (80% power, $\alpha=0.05$). Data were analyzed in R Studio (R version 4.4.3) using “tidyverse”, “ggplot2”, “stats”, “nortest”, “did”, and “vegan” packages. To illustrate the central distribution trends continuous variables are presented as mean $\pm$ SD and median (Me) with IQR. Non-parametric tests (Mann–Whitney U, Wilcoxon, χ^2 , McNemar) were used, respectively, for dependent and independent features. Difference-in-Differences (DiD) regression and PERMANOVA were applied for longitudinal and compositional data. The DiD model considered the level of leukocytes in urine in the preoperative and postoperative periods for patients in each of the study groups as a target variable. Statistical significance was set at $p < 0.05$.

Results

Ninety-five patients (62 men and 33 women, mean age 48.8 ± 13.0 years) completed the study. Clinical and demographic characteristics, including age, sex, stone size, density, operation time and preoperative laboratory parameters, were comparable between the control (n=48) and dioxidine (n=47) groups (all $p > 0.1$; Table 1).

Table 1. Clinical and demographic characteristics of patients in both groups before surgery (median (Me), interquartile range (IQR), mean $\pm$ SD).

Sign	Group I, n=48	Group II, n=47	Statistics, p - the level of differences
Age, years	Me=47.5; IQR: 37 -61.5 49.7 $\pm$ 15,2	Me=49; IQR: 39,5 -55,5 47.8 $\pm$ 12,1	W=1192, $p = 0.6363$
Gender distribution: M/W	21/27	12/35	$\chi^2=3,477$, $p=0,063$
Hemoglobin before surgery, g/l	Me=132; IQR: 121 -147,5 135,1 $\pm$ 15,5	Me=137; IQR: 124 -147,5 136,6 $\pm$ 14,6	W=1029.5, $p = 0.4655$
Creatinine in blood before surgery, mmol/l	Me=86,1; IQR: 75,3 -100,9 90,1 $\pm$ 23,0	Me=91,7; IQR: 77,2 -110,0 97,9 $\pm$ 25,5	W = 943, $p = 0.1696$
White blood cells in the blood before surgery, $10^9/l$	Me=7,11; IQR: 5,98 -8,5 7,16 $\pm$ 1,5	Me=7,5; IQR: 6,2 -8,5 7,3 $\pm$ 1,6	W = 1066, $p = 0.6468$
White blood cells in urine before surgery, units per unit.	Me=8,5; IQR: 4,0 -13,25 10,4 $\pm$ 9,4	Me=9,0; IQR: 6,0 -14,0 13,02 $\pm$ 11,7	W = 977.5, $p = 0.2633$
Ph in urine before surgery, units.	Me=1017; IQR: 1015 -1020 1017 $\pm$ 4,6	Me=1018; IQR: 1014 -1022 1019 $\pm$ 5,8	W = 1022, $p = 0.4297$
Operation Time (mean $\pm$ Std) min	66.6 $\pm$ 20	63.7 $\pm$ 14	$p > 0.05$
Stone Size (mean $\pm$ Std) mm	26 $\pm$ 10	30 $\pm$ 12	$p > 0.05$

As shown in Table 2, there were no significant intergroup differences in postoperative hemoglobin, serum creatinine, or blood leukocyte levels (all $p > 0.1$). At the same time, intracavitary dioxidine administration resulted in a marked and sustained anti-inflammatory effect. One week after PCNL, the median urine leukocyte count was 9.1 (IQR 5.0–10.25) in the control group versus 2.45 (IQR 0.0–5.0) in the dioxidine group ($p < 0.001$). This advantage persisted at the 3-month follow-up: 5.8 (IQR 2.25–10.0) versus 1.87 (IQR 0.0–4.0), respectively ($p < 0.001$; Table 2).

Within-group comparisons demonstrated a significant postoperative decrease in hemoglobin levels in both groups ($p < 0.001$). Blood leukocyte count increased significantly only in the control group ($p = 0.011$). The dynamics of these parameters are illustrated in box plots (Figure 1).

The most pronounced positive changes were observed in urinary leukocyturia in the dioxidine group. In the control group, urine leukocyte count did not

decrease significantly at 1 week ($p = 0.240$), while in the dioxidine group it decreased markedly ($p < 0.001$) and remained low at 3 months ($p < 0.001$). These dynamics are presented in Figure 2. Changes in urine pH are shown in Figure 3.

Analysis of kidney stone size dynamics (Figure 4) showed a significant reduction in both groups one week after surgery ($p < 0.01$). However, at 6 months, median stone size in the control group returned toward preoperative values ($p > 0.1$), whereas in the dioxidine group it remained significantly lower than baseline ($p < 0.001$), with a significant intergroup difference ($p < 0.001$).

Stone composition analysis (Table 3) revealed no significant differences between the groups in the proportion of individual components or frequency of stone types (all $p > 0.1$). PERMANOVA confirmed the absence of differences in overall stone structure ($p = 0.499$).

No adverse reactions to dioxidine were recorded. Clinically significant complications (Clavien–Dindo

Table 2. Comparison of clinical and laboratory parameters in both groups after surgery (Me, IQR, mean $\pm$ SD).

Sign	Group I, n=48	Group II, n=47	Statistics, p - the level of differences
Hemoglobin after surgery, g/l	Me=126; IQR: 115 -140 126,8 $\pm$ 16,3	Me=133; IQR: 119 -141,5 130,8 $\pm$ 14,8	W=958,5, p = 0.208
Creatinine after surgery, mmol/l	Me=87,3; IQR: 74,3 -100,3 82,6 $\pm$ 19,5	Me=87,9; IQR: 75,6 -105,3 99,5 $\pm$ 22,9	W=955,5, p = 0.2004
White blood cells after surgery, 10 ⁹ /l	Me=7,5; IQR: 6,88 -8,93 7,89 $\pm$ 1,8	Me=7,2; IQR: 6,0 -8,0 7,3 $\pm$ 1,5	W=1337,5, p = 0.1191
White blood cells in urine after surgery (1 week), units per unit.	Me=7,0; IQR: 5,0 -10,25 9,1 $\pm$ 6,7	Me=0,0; IQR: 0,0 -5,0 2,45 $\pm$ 3,5	W = 1857,5 p< 0.001***
White blood cells in urine after surgery (3 months), units per unit.	Me=5,0; IQR: 2,25 -10,0 5,8 $\pm$ 4,5	Me=0,0; IQR: 0,0 -4,0 1,87 $\pm$ 2,3	W = 1727, p< 0.001***
pH in urine after surgery (1 week), units.	Me=1016; IQR: 1015 -1018 1016 $\pm$ 4,0	Me=1018; IQR: 1013 -1022 1018 $\pm$ 5,2	W = 967.5, p = 0.2313
pH in urine after surgery (3 months), units.	Me=1016; IQR: 1015 -1019 1017 $\pm$ 4,6	Me=1017; IQR: 1013 -1020 1017 $\pm$ 4,7	W = 1200,5 p = 0.5901

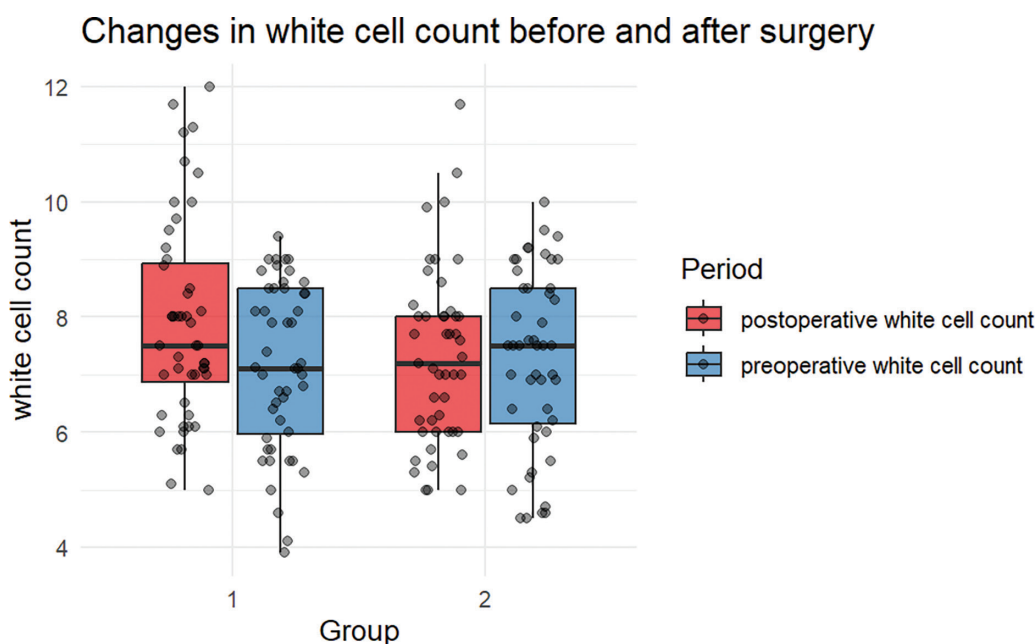


Figure 1. Box plots of blood leukocyte levels before and after surgery for Group I and Group II.

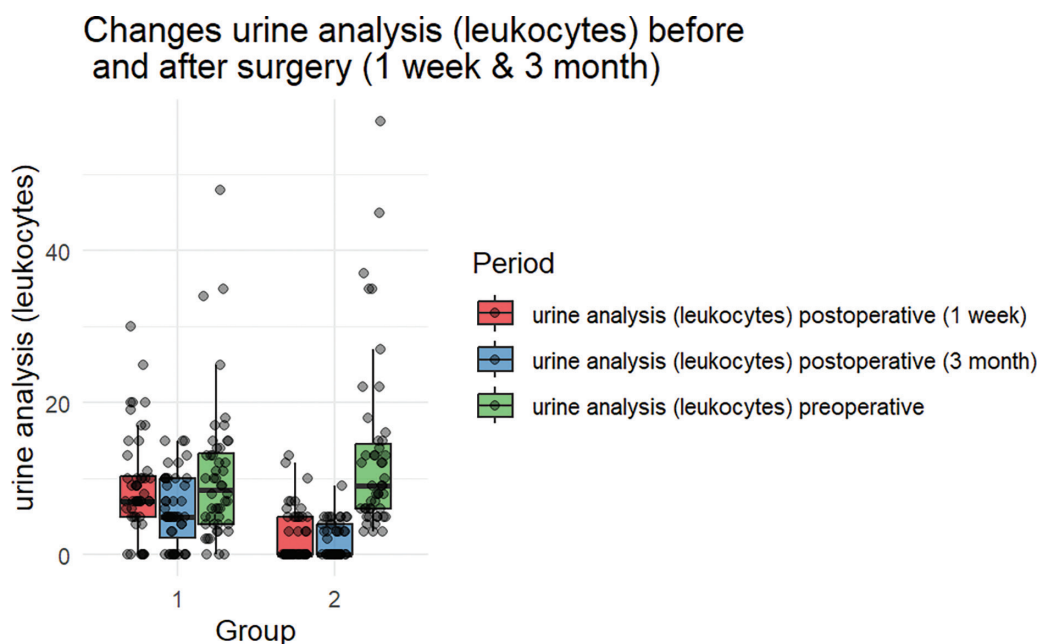


Figure 2. Box plots of urine leukocyte count before surgery, at 1 week, and at 3 months for Group I and Group II.

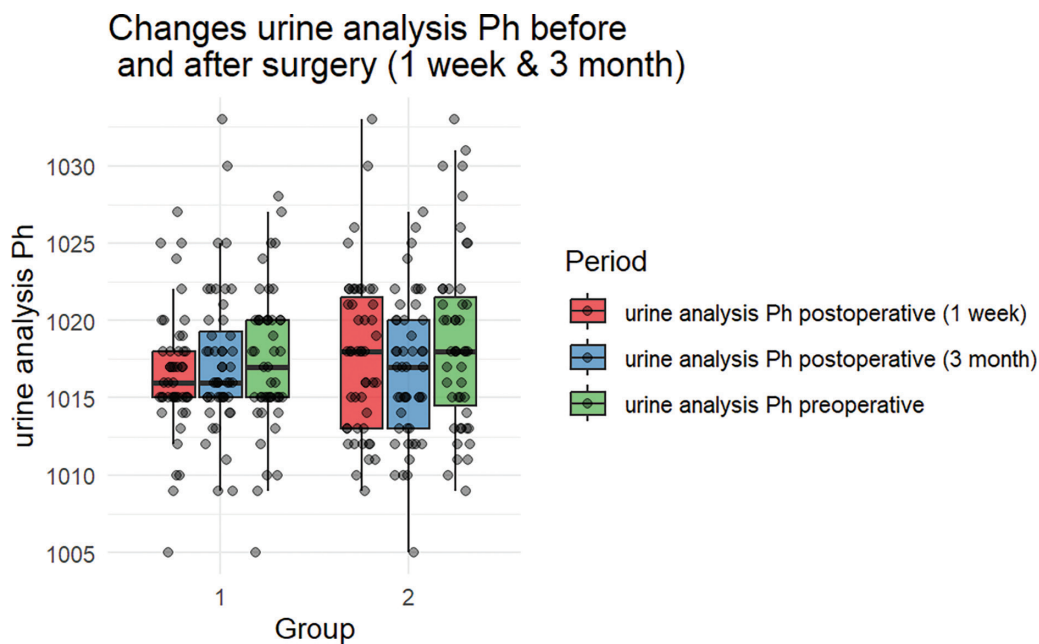


Figure 3. Box plots of urine pH before surgery, at 1 week, and at 3 months for Group I and Group II.

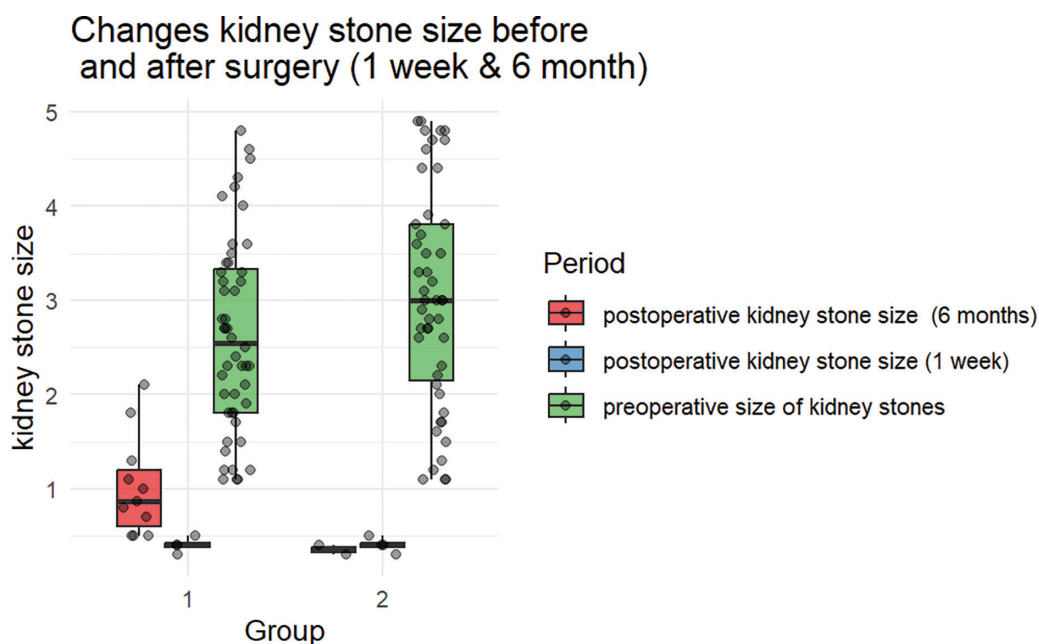


Figure 4. Box plots of kidney stone size before surgery, at 1 week, and at 6 months for Group I and Group II.

Table 3. Intragroup comparison of clinical and laboratory parameters of both groups of patients before and after surgery (median (Me) and interquartile range (IQR), mean± standard deviation).

Sign	Group I, n=48	Group II, n=47	Statistics, p-the level of differences
Weddellite	Me=35%; IQR: 22,5% -52,5%	Me=35%; IQR: 22,8% -56%	$W = 77,0$ $p=1$
	Quantity 11	Quantity 14	$\chi^2=0,578$, $p=0,448$
Whewellite	Me=41,5%; IQR: 21,5% -72,3%	Me=50%; IQR: 30,5% -70%	$W = 297,0$ $p=0,5462$
	Quantity 30	Quantity 22	$\chi^2=2,360$, $p=0,125$
Uric acid	Me=80%; IQR: 30% -80,5%	Me=75%; IQR: 57,5% -77,5%	$W = 36,5$, $p= 0.6313$
	Quantity 9	Quantity 7	$\chi^2=0,252$, $p=0,616$
Struvite	Me=80%; IQR: 60% -90%	Me=74%; IQR: 57,5% -86,3%	$W = 36,0$, $p= 0.6011$
	Quantity 13	Quantity 12	$\chi^2=0,029$, $p=0,864$

Table 3 (Continued)

Sign	Group I, n=48	Group II, n=47	Statistics, p-the level of differences
Ammonium urate	<i>Me</i> =60%; <i>IQR</i> : 30% -81%	<i>Me</i> =58%; <i>IQR</i> : 50% -100%	<i>W</i> = 18,5, <i>p</i> = 0,9351
	Quantity 5	Quantity 7	$\chi^2=0,431$, <i>p</i> =0,512
Brushite	<i>Me</i> =65%; <i>IQR</i> : 56,3% -77,5%	<i>Me</i> =82,5%; <i>IQR</i> : 58,8% -100%	<i>W</i> = 7,0 <i>p</i> = 0,8824
	Quantity 4	Quantity 4	$\chi^2=0,001$, <i>p</i> =0,976
Carbonatapatite	<i>Me</i> =35%; <i>IQR</i> : 26% -53,5%	<i>Me</i> =35%; <i>IQR</i> : 28% -55%	<i>W</i> = 7,5 <i>p</i> = 0,8614
	Quantity 23	Quantity 21	$\chi^2=0,100$, <i>p</i> =0,752

Table 4. Species-specific paired comparison of bacterial flora in preoperative bladder urine versus intraoperative pelvic urine in Group I and Group II.

A type of bacterium	Group I, n=48			Group II, n=47		
	Before the operation	During the operation	P-уровень	Before the operation	During the operation	P-level
ecoli	8	4	0,102	10	7	0,257
	Coincidences 3			Coincidences 5		
staph	2	9	0,020*	1	7	0,015*
	Coincidences 1			Coincidences 1		
candida	4	2	0,318	1	0	0,318
	Coincidences 1			Coincidences 0		
pseudo	1	3	0,318	1	2	0,318
	Coincidences 0			Coincidences 1		
klebsiel	1	8	0,020*	1	5	0,046*
	Coincidences 0			Coincidences 1		
entero	3	0	0,084	3	1	0,158
	Coincidences 0			Coincidences 1		
proteus	2	8	0,034*	2	6	0,103
	Coincidences 1			Coincidences 1		
strep	1	0	0,318	0	1	0,318
	Coincidences 0			Coincidences 0		

grade $\geq$ II) occurred in 25.0% of patients in the control group and 4.3% in the dioxidine group ($p = 0.004$). Stone recurrence or residual fragments at 6 months were observed in 22.9% and 4.3% of patients, respectively ($p = 0.008$).

Comparison of structural stone composition between groups species-specific analysis (Table 4) revealed important differences in the frequency of detection of several uropathogens. In the control group, statistically significant increases in the intraoperative

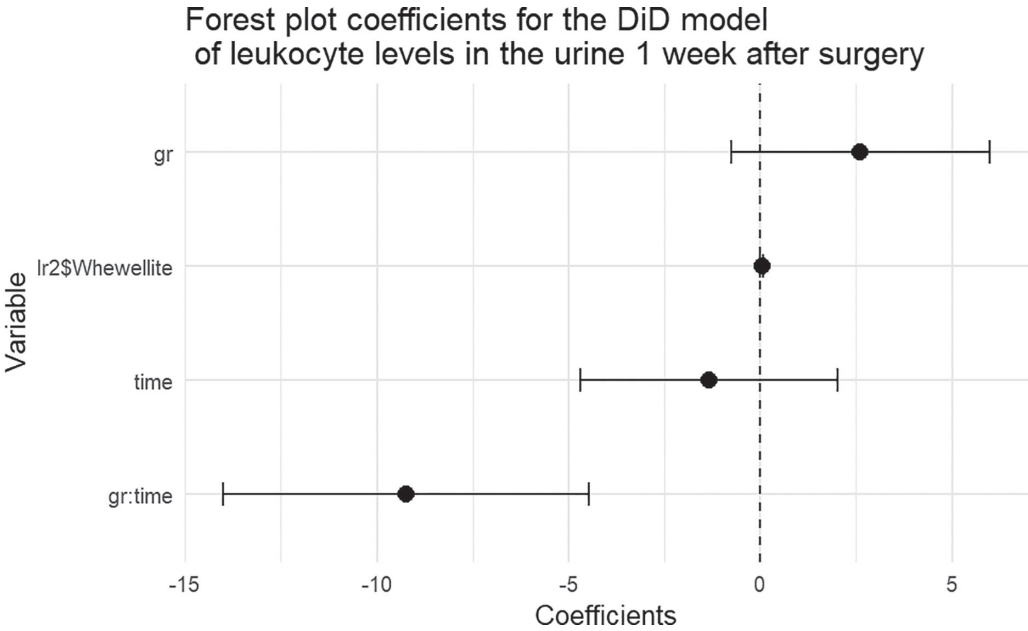


Figure 5. Forest plot of the effect of the proposed method on urine leukocytes at 1 week after surgery.

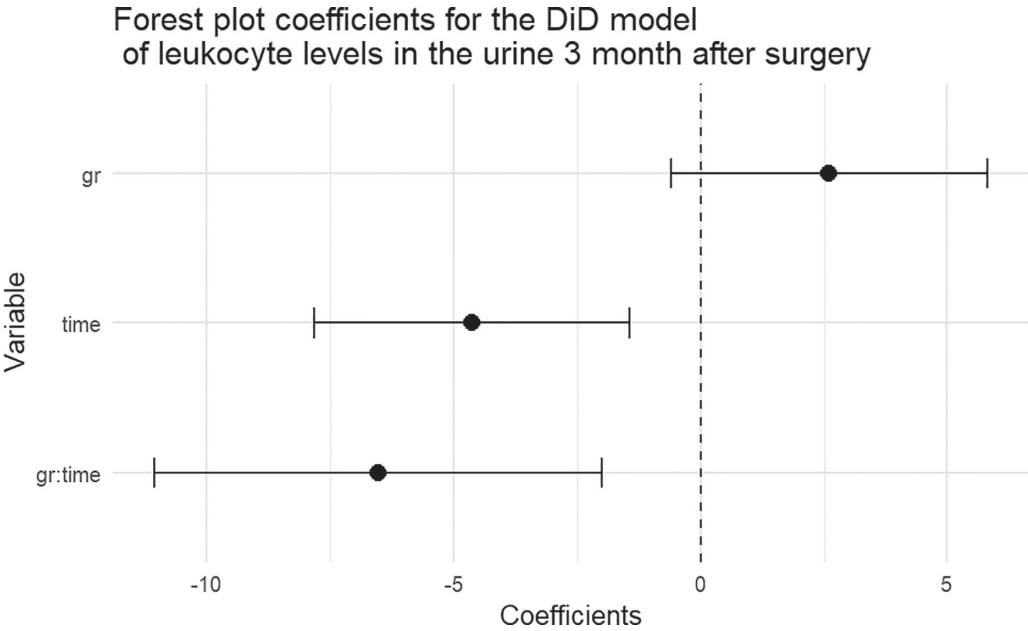


Figure 6. Forest plot of the effect of the proposed method on urine leukocytes at 3 months after surgery.

pelvic urine were observed for *Staphylococcus* spp. ($p = 0.020$), *Klebsiella* spp. ($p = 0.020$), and *Proteus* spp. ($p = 0.034$). In the dioxidine group, significant increases were noted for *Staphylococcus* spp. ($p = 0.015$) and *Klebsiella* spp. ($p = 0.046$).

Overall concordance analysis demonstrated statistically significant differences between bladder and pelvic urine flora (Group I: $\chi^2 = 3.930$, $p = 0.048$; Group II: $\chi^2 = 4.481$, $p = 0.035$). These findings indicate that intraoperative sampling from the renal pelvis detects a substantially higher rate of bacterial colonization than standard preoperative bladder urine culture. This highlights the presence of occult infection within the collecting system in patients with nephrolithiasis, which may not be identified by routine bladder urine testing alone.

Difference-in-Differences (DiD) regression models confirmed a strong independent effect of the intervention. Intracavitary dioxidine reduced urine leukocyte count by 9.24 ± 2.42 per high-power field at 1 week ($p < 0.001$) and by 6.52 ± 2.29 per hpf at 3 months ($p = 0.005$), independent of stone composition. These results are illustrated in forest plots (Figures 5,6).

Discussion

This randomized trial demonstrated that intracavitary administration of dioxidin as an adjunct to standard antibiotic therapy during PCNL significantly reduced postoperative infectious complications, bacterial persistence, and residual stone rates in patients with nephrolithiasis. Although both groups had comparable baseline characteristics and operative parameters, the addition of local dioxidin irrigation yielded superior short- and mid-term microbiological outcomes without increasing adverse events.

The prevention of postoperative infection following PCNL remains a major clinical concern, particularly in patients with struvite or mixed-infectious stones, where bacterial colonization within the stone matrix leads to recurrent infection and stone regrowth despite negative urine cultures (10, 11). Our findings align with prior evidence that infection control requires not only systemic but also local antimicrobial activity at the surgical site. Previous studies have emphasized

that bacterial biofilms within the collecting system are poorly penetrated by conventional antibiotics, leading to persistent bacteriuria and a higher rate of sepsis or recurrence (12). Intracavitary antiseptic administration, therefore, offers a mechanistic advantage by directly targeting residual pathogens within the renal pelvis and stone fragments.

Dioxidin, a hydroxymethylquinoxalinedioxide derivative, exerts potent bactericidal effects against Gram-negative and multidrug-resistant organisms, including *Pseudomonas aeruginosa* and *Klebsiella* spp., which are common etiologic agents of postoperative infections (7). Recent studies have renewed interest in its local application, reporting favorable safety profiles and reduced infection-related morbidity after urological and general surgical procedures (13,14). In our series, dioxidin use resulted in a nearly fivefold reduction in culture positivity at 3 months and a substantial decrease in clinically relevant complications. These findings support the hypothesis that local antimicrobial prophylaxis can effectively disrupt bacterial colonization and biofilm formation after stone surgery.

Notably, the residual stone rate was significantly lower in the dioxidin group, suggesting that improved infection control may facilitate better stone clearance and prevent early recurrence. This observation is consistent with the pathophysiological link between infection and stone regrowth, wherein bacterial urease activity promotes crystallization and matrix formation (15,16). By reducing microbial load, dioxidin may indirectly enhance the durability of stone-free outcomes following PCNL.

Dioxidine has been used for many years in several CIS countries for the treatment of severe purulent infections due to its broad-spectrum bactericidal activity against Gram-negative and multidrug-resistant pathogens (7). Although concerns regarding potential systemic toxicity exist in the literature, primarily associated with high-dose intravenous administration, the present study employed low-dose intracavitary irrigation (1% solution, 10 ml). This route of administration minimizes systemic absorption while achieving high local concentrations in the pelvicalyceal system. Importantly, no local or systemic adverse effects, including methemoglobinemia or deterioration of renal function (serum creatinine levels remained stable in both groups, all $p > 0.05$), were observed.

Although patients receiving dioxidine demonstrated a lower rate of residual or recurrent stones during follow-up, this finding should not be interpreted as evidence of improved surgical stone clearance. Stone-free status after PCNL is influenced by multiple factors, including stone burden, anatomical complexity, operative time, access characteristics, and surgeon experience. In the present study, baseline stone size, stone density, and operative time were comparable between the study groups. Nevertheless, validated nephrolithometry scoring systems (e.g., the S.T.O.N.E. nephrolithometry score or Guy's Stone Score) were not prospectively assessed, and therefore residual confounding cannot be completely excluded. It is plausible that enhanced local infection control with dioxidine reduced bacterial persistence and bio-film formation on microscopic residual fragments, thereby lowering the risk of early infection-related stone regrowth rather than directly affecting surgical stone clearance.

Several limitations should be acknowledged. The sample size was modest and limited to a single center, and long-term outcomes beyond 6 months were not assessed. Nevertheless, the randomized design and consistent surgical and postoperative protocols strengthen the validity of these results.

Adjunctive intracavitary dioxidine administration during PCNL appears to provide effective local infection control, reduce postoperative bacterial persistence, and lower residual stone rates without additional risk. Larger multicenter studies with extended follow-up are warranted to validate its role as a complementary measure in the meta-phylaxis of infectious stones.

Conclusions

This randomized controlled trial demonstrated that adjunctive intracavitary administration of 1% dioxidine during and after percutaneous nephrolithotomy is a safe and effective strategy for reducing postoperative infectious complications in patients with large kidney stones. Compared with standard

systemic antibiotic therapy alone, local dioxidine irrigation significantly decreased bacteriuria, leukocyturia, clinically significant complications (Clavien-Dindo grade $\geq$ II), and stone recurrence at 6 months, without compromising renal function or increasing adverse events.

The results highlight the limitations of standard preoperative bladder urine cultures and the superior diagnostic value of intraoperative pelvic urine sampling for detecting occult infection within the collecting system. Difference-in-Differences analysis confirmed a robust independent effect of the intervention on key inflammatory markers, independent of stone composition.

These findings support the broader incorporation of local antiseptics as a complementary measure in the metaphylaxis of nephrolithiasis. Larger multicenter randomized trials with longer follow-up are warranted to confirm the long-term benefits and establish the optimal role of dioxidine in contemporary endourological practice.

Ethic approval: Local Ethics Committee of West Kazakhstan Marat Ospanov Medical University, Protocol No. 23, December 12, 2023.

Conflict of interest: Each author declares that he or she has no commercial associations (e.g. consultancies, stock ownership, equity interest, patent/licensing arrangement etc.) that might pose a conflict of interest in connection with the submitted article.

Authors' contribution: Imzharov TA - Conceptualization, data collection, manuscript preparation. Balci M - Data analysis. Zhakiev BS - Data collection, manuscript review. Kurmangaliyev OM - Data collection. Sarkulov MN - Data collection, manuscript review. Aslan Y - Data analysis, expert consultation. Pavlov VN - Study supervision, manuscript review. Lakman IA - Data analysis. Kelimberdiev MS - Data analysis, manuscript preparation. All authors have read and approved the final version of the manuscript.

Declaration on the use of AI: AI was used for language editing and reference style checking.

Consent for publication: Informed consent was obtained from all subjects involved in this study.

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